

Early Social Enrichment Shapes Social Behavior and Nerve Growth Factor and Brain-Derived Neurotrophic Factor Levels in the Adult Mouse Brain

Igor Branchi, Ivana D'Andrea, Marco Fiore, Veronica Di Fausto, Luigi Aloe, and Enrico Alleva

Background: Early experiences produce persistent changes in brain and behavioral function. We investigate whether being reared in a communal nest (CN), a form of early social enrichment that characterizes the natural ecological niche of many rodent species including the mouse, has effects on adult social/aggressive behavior and nerve growth factor (NGF) and brain-derived neurotrophic factor (BDNF) levels in mice.

Methods: The CN consisted of a single nest where three mothers kept their pups together and shared care-giving behavior from birth to weaning (postnatal day 25).

Results: Compared to standard laboratory conditions, in CN condition, mouse mothers displayed higher levels of maternal care. At adulthood, CN mice displayed higher propensity to interact socially and achieved more promptly the behavioral profile of either dominant or subordinate male. Furthermore, CN adult mice showed higher NGF levels, which were further affected by social status, and higher BDNF levels in the brain.

Conclusions: Our findings indicate that CN, a highly stimulating early social environment, produces differences in social behavior later in life associated with marked changes of neurotrophin levels in selected brain areas, including hippocampus and hypothalamus.

Key Words: Communal nesting, neurotrophins, maternal behavior, aggressive behavior, psychiatric diseases, early experiences

During early postnatal development, several important processes that shape the mammalian brain are taking place. This highly plastic period characterized by neuronal differentiation, maturation, and synaptogenesis offers the possibility for epigenetic factors to affect brain structure and function (Rosenzweig 2003).

An intact, natural-type, early social environment is crucial for brain and behavioral development, as shown by the disrupting effects of its impoverishment or deterioration (Canetti et al 1997; Harlow 1958; Pollak 2003; Rutter et al 1997; Trickett and McBride-Chang 1995). Indeed, children who experience severe perturbations in care are at higher risk for the emergence of behavioral problems, including social behavior (Dodge et al 1995; Rogosch et al 1995) or of psychiatric diseases later in life (Dube et al 2001; Nawa et al 2000). Manipulations of the mother-infant interaction in rodents have consequences on behavioral and neuroendocrine responses that persist for the entire life-span (Cirulli et al 2003; Levine 1957; Meaney 2001). For instance, the offspring of rodent mothers exhibiting less care-giving behavior show, at adulthood, an increased response to stressful situations (Liu et al 1997), producing high levels of adrenal steroids that exert disrupting effects on the nervous system (McEwen 1998).

Neurotrophins, particularly nerve growth factor (NGF) and brain-derived neurotrophic factor (BDNF), are key molecules in neuronal proliferation and differentiation and in synaptogenesis

(McAllister et al 1999) that have been hypothesized to mediate the effects of experience on brain structure and function (Branchi et al 2004). A series of experiments found that manipulations of the early social environment exert both short- and long-term effects on brain neurotrophin expression in both mice and rats (Cirulli et al 2003; Liu et al 2000). A daily 3-hour maternal separation from postnatal day (PND) 2 to PND 14 induces, on PND 17, an increase of NGF and BDNF messenger-RNA expression in the hippocampus and prefrontal cortex of rats (Cirulli et al 2003; Roceri et al 2004) and, at adulthood, a selective BDNF reduction in the prefrontal cortex (Roceri et al 2004). In humans, altered neurotrophin levels have been widely related to several psychiatric disorders, suggesting a role for neurotrophins, particularly BDNF, in the onset and progression of pathologies such as schizophrenia and depression (Castren 2005; Nawa et al 2000; Schumacher et al 2005).

The aim of the present work was to characterize the role of early social environment on adult behavior. Communal nesting (CN), the sharing of maternal care-giving behavior among multiple females in a single nest, characterizes the natural ecological niche of developing pups in many rodent species and represents a form of early social enrichment when compared with standard laboratory rearing conditions (Hayes 2000; Sayler and Salmon 1971). We therefore examined the effects of being reared in CN—consisting in three mothers keeping their pups together and sharing care-giving behavior in a single nest—on social/aggressive behavior of offspring at adulthood. Because neurotrophin levels are reportedly modulated by social factors (Aloe et al 1986; Branchi et al 2004; Cirulli et al 2003), we also evaluated whether CN produces long-term changes of NGF and BDNF levels in different brain areas.

Methods and Materials

Animals and Breeding Procedures

Twenty-four male and 48 female mice of an outbred CD-1, Swiss-derived strain (ICR), weighing 25–27 g, were purchased from a commercial breeder (Harlan, Correzzana, Milan, Italy). Upon arrival at the laboratory, the animals were housed in an air-conditioned room (temperature $21 \pm 1^\circ\text{C}$, relative humidity $60 \pm 10\%$) with lights on from 8:00 PM to 8:00 AM. Males and females were housed in same-gender groups of six individuals in $42 \times 17 \times 14$

From the Section of Behavioural Neurosciences (IB, ID, EA), Department of Cell Biology and Neuroscience, Istituto Superiore di Sanità; and the Institute of Neurobiology and Molecular Medicine (MF, VDF, LA), Consiglio Nazionale delle Ricerche, Rome, Italy.

Address reprint requests to Igor Branchi, Section of Behavioural Neurosciences, Department of Cell Biology and Neuroscience, Istituto Superiore di Sanità, Viale Regina Elena 299, I-00161 Roma, Italy; E-mail: branchi@iss.it.

Received October 7, 2005; revised January 2, 2006; accepted January 3, 2006.

cm Plexiglas boxes with metal tops and sawdust as bedding and with pellet food (enriched standard diet purchased from Mucedola, Settimo Milanese, Italy) and tap water *ad libitum*.

All animal handling and experimental procedures were performed according to European Communities guidelines (EC Council Directive 86/609) and the Italian legislation on animal experimentation (Decreto L.vo 116/92).

Nesting Conditions

After 1 week of acclimatization to the animal room, mice were assigned to one of the two experimental groups: standard nesting (SN) or CN. For the SN group, 12 males and 12 females were used to form breeding pairs housed in $33 \times 13 \times 14$ cm Plexiglas boxes. For the CN group, 12 males and 36 females were used to form breeding groups, made up of 1 male and 3 females, housed in $42 \times 17 \times 14$ cm Plexiglas boxes. In both cases, at approximately pregnancy day 16 the males were removed from the breeding cage. The litters were culled on PND 1 (birth = PND 0) to six males and two females. In the CN group, in each female trio, there was a discrepancy of up to 6 days in the age of the pups. Pups were weaned on PND 25, and males of each litter were housed in groups of six animals in $42 \times 17 \times 14$ cm Plexiglas boxes. Females were not used in this study.

Maternal Care Observations

Eight SN and eight CN cages were observed on PNDs 4, 6, 8, 10, 12, and 14. Maternal behavior was scored during six sessions each day. Data were collected in each session for each cage, with one-zero sampling, over 25 20-sec observations that were 140 sec apart. The observer recorded whether the behavior was present or not during the 20-sec observation; more than one behavior could be present and thus recorded during the same 20-sec observation. The sessions started at 6:00 PM, 9:00 PM, 12:00 AM, 3:00 AM, 6:00 AM, and 9:00 AM. The first and last sessions were during the dark phase of the 12:12 cycle and were performed under dim red light illumination.

Maternal behaviors scored were: no dams in the nest: self-explanatory; number of dams in the nest: self-explanatory; blanket nursing: the dam is over the pups, relatively immobile, bilaterally symmetrical, with the head not depressed, and in a low dorsal arch posture supported by rigid forelimbs and hind limbs (low kyphosis) or in a low dorsal arch posture supported by rigid forelimbs or rigid hind limbs (partial kyphosis) or lying flat on top of the pups with little or no limb support; passive nursing: the dam is lying down on her side with more than one pup usually attached to the nipples; arched-back nursing: the dam is immobile and in a high upright dorsal arch posture supported by rigid forelimbs and hind limbs, the head is depressed, the trunk and limbs are bilaterally symmetrical, and pups are attached to the nipples; total nursing is the sum of the three nursing positions; licking: licking of the pup body; stepping: the dam steps gently over the pup body and stimulate them; retrieval: with the incisors, the dam picks up the pup gently by its dorsal skin and carries it to the nest.

Social Interaction Test

From PND 240 to PND 245 (8 months of age), after 3 weeks of isolation (single-housed) in a $33 \times 13 \times 14$ cm Plexiglas box, mice underwent a social interaction test. On each day of 5 consecutive days, experimental subjects underwent a social interaction session consisting of being placed for 20 min in a neutral cage, identical to the home cage, with a standard opponent, an adult CD-1 male of the same weight that had been previously isolated. The same experimental animal was paired with the same standard opponent throughout the 5 days of test. Standard opponents were marked by

a blue, scentless, and nontoxic felt pen 30 min before testing, to discriminate the experimental subject from the standard opponent during data collection. All tests were carried out between 9:30 AM and 3:30 PM (i.e., during the red-light period), and animals were transferred to the experimental room at least 30 min before the test to let them acclimatize to the test environment.

During the agonistic interaction, behavior was video-recorded with a digital video camera (Model TR 7000E, Sony, Tokyo). The behavioral analysis was carried out by replaying videotapes and collecting data with the use of commercial software (Observer 3.0, Wageningen, The Netherlands). All scores were assigned from the same observer, who was unaware of the group assignment of the animals.

The behavioral categories and elements scored were:

- Agonistic behaviors: latency to the first attack, the time from the beginning of the test session to the first biting attack; attack, fighting episode when a mouse belonging to the experimental group approaches the opponent and bites it; aggressive grooming, violent grooming of the animal on the back of the partner; chase, the animal goes after the opponent; offensive upright posture, the animal stands on its hind limbs facing the opponent aggressively; defensive upright posture, the animal stands on its hind limbs and pushes the aggressive opponent with its forepaws; crouched posture, the animal lies on its ventrum, the head flat on the cage floor (this posture is often observed as an answer to aggressive grooming performed by the partner); submissive upright posture, the animal stands on its hind limbs, the head pulled far back and its body rigid; freezing, animal reacts to the physical movements of the partner by remaining motionless; fleeing, animal moves very rapidly away from the partner, often associated with screaming; tail rattling, fast tail vibration often observed in a distance ambivalence situation
- Affiliative behaviors: anogenital sniff, sniffing the anogenital area of the partner; nose sniff, sniffing the head and the snout region of the partner; allogrooming, grooming the partner; body sniff, sniffing any other area of the body of the partner
- Non-social activities: wall rearing, animal stands on its hind limbs and touches the walls of the cage with the forelimbs; bar holding, animal grasps the metal top of the cage holding itself above the level of the ground; self-grooming, animal licks and mouths its own fur, sometimes helping itself with its forepaws; digging, animal digs the sawdust with the forelimbs, often kicking it away with the hind limbs.

At the end of the last fighting session, each experimental subject was assigned to either the dominant or the subordinate group: a mouse was scored as having achieved the dominant status when, in the course of the fifth encounter, it displayed a consistent amount of attacks without being attacked by its opponent. Conversely, a mouse was considered subordinate when it was continuously attacked and defeated by the opponent, without attacking back, and showing fully defensive and submissive behavior (Alleva 1993; Pietropaolo et al 2004).

Tissue Dissection

Animals were euthanized with an overdose of pentobarbital (Carlo Erba Reagenti, Milan, Italy); the brain was quickly removed, and the hippocampal, striatal, and hypothalamic tissues were dissected with a mouse brain matrix (ASI Instruments, Warren, Michigan) and stored at -70°C until used. Tissues were then homoge-

nized and centrifuged at 8500 rpm, and the supernatant was used for NGF and BDNF analyses.

NGF Determination

The NGF evaluation was carried out in the hippocampus, striatum, hypothalamus, and serum with an enzyme-linked immunosorbent assay kit (NGF Emaxtm ImmunoAssay System number G7631, Promega, Madison, Wisconsin) following the instructions provided by the manufacturer. Tissues were homogenized in the kit calibration buffer and centrifuged. The brain tissues were homogenized with ultrasonication in extraction buffer .2% triton. Briefly, 96-well immunoplates were coated with 100 μ L per well of monoclonal anti-mouse NGF antibody. After an overnight incubation at 4°C, the plates were washed three times with wash buffer and the samples were incubated in the coated wells (100 μ L each) for 2 hours at room temperature with shaking. After an additional five washes, the immobilized antigen was incubated with an anti-human NGF antibody for 2 hours at room temperature with shaking. The plates were washed again with wash buffer and then incubated with an anti-IgY horseradish peroxidase (HRP) for 1 hour at room temperature. After another wash the plates were incubated with a tetramethylbenzidine (TMB)/peroxidase substrate solution for 15 min, and then phosphoric acid 1 mol/L (100 μ L/well) was added to the wells. The colorimetric reaction product was measured at 450 nm with a microplate reader (Dynatech MR 5000, Dynatech Laboratories, Chantilly, Virginia). The NGF concentrations were determined from the regression line for the NGF standard (ranging from 7.8 to 500 pg/mL purified mouse NGF) incubated under similar conditions in each assay. Under these conditions, the recovery of NGF in our assay ranged from 80% to 90%. The sensitivity of the assay was about 3 pg/g of wet tissue, and cross-reactivity with other related neurotrophic factor (BDNF, Neurotrophin-3, and Neurotrophin-4) was less than 3%. All assays were performed in triplicate.

BDNF Determination

The concentrations of BDNF in the same brain areas of the NGF assay were measured by an enzyme-linked immunosorbent assay kit (BDNF Emaxtm ImmunoAssay System number G6891, Promega) according to the instructions of the manufacturer. Tissues were homogenized in the kit calibration buffer and centrifuged. The brain tissues were homogenized with ultrasonication in extraction buffer .2% triton. Briefly, 96-well immunoplates were coated with 100 μ L per well of monoclonal anti-mouse BDNF antibody. After an overnight incubation at 4°C, the plates were washed three times with wash buffer and the samples were incubated in the coated wells (100 μ L each) for 2 hours at room temperature with shaking. After an additional five washes the immobilized antigen was incubated with an anti-human BDNF antibody for 2 hours at room temperature with shaking. The plates were washed again with wash buffer and then incubated with an anti-IgY HRP for 1 hour at room temperature. After another wash the plates were incubated with a TMB/peroxidase substrate solution for 15 min, and then phosphoric acid 1 mol/L (100 μ L/well) was added to the wells. The colorimetric reaction product was measured at 450 nm with a microplate reader (Dynatech MR 5000, PBI International). The BDNF concentrations were determined from the regression line for the BDNF standard (ranging from 7.8 to 500 pg/mL purified mouse BDNF) incubated under similar conditions in each assay. The sensitivity of the assay was about 15 pg/g of BDNF, and cross-reactivity with other related neurotrophic factors (NGF, Neurotrophin-3, and Neurotrophin-4) was less than 3%. All assays were performed in duplicate.

Statistical Analysis

All data were analyzed by analyses of variance, with the nesting condition (standard vs. communal) and social status (dominant vs. subordinate) as between-subject variables. Subject was considered as random factor nested within nesting condition and social status; time (postnatal day or daily sessions) as repeated measures for subjects. Post-hoc comparisons were performed with Tukey Honestly Significant Difference (HSD) test between groups, differing for only one factor (i.e., either nesting condition or social status). Comparisons between groups differing for both factors are meaningless, because it would not be possible to identify the relative contribution of each feature to an observed difference. Statistical analyses were carried out with the software Statview II (Abacus Concepts, Berkeley, California).

Results

Higher Levels of Maternal Behavior in CN

We found, in agreement with previous works (Sale et al 2004; Saylor and Salmon 1971), increased levels of maternal care provided to CN pups (Figure 1). Maternal behavior is interposed by periods during which the mother is out of the nest and pups remain alone. We therefore scored the time spent by pups alone in the nest and found that it was significantly lower in CN compared with SN condition [$F(1,14) = 81.337, p < .0001$]. The CN pups were alone less, because often a mother was replaced by another when leaving the nest. Furthermore, the number of dams in the nest when the nest was attended was higher in CN than in SN condition [$F(1,14) = 132.544, p < .0001$].

The CN mothers displayed higher total levels of nursing compared with SN ones [$F(1,14) = 128.516, p < .0001$]. Different types of nursing positions were adopted for different durations by CN compared with SN mothers, probably owing to practical constraints

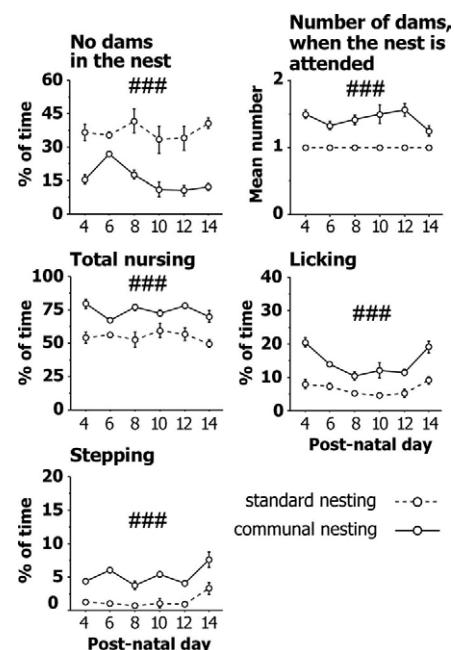


Figure 1. Maternal behavior in communal nesting (CN) and standard nesting (SN) condition: higher levels of maternal care were performed in CN condition. Data were analyzed as the percentage of observations (150 observations per animal per day) in which animals were observed to be engaged each behavior analyzed. *** $p < .001$, CN versus SN mice; $n = 8$ for each group. Data are means $\pm$ SEM.

related to the need of lactating a different number of pups. The CN mothers performed more blanket nursing [$F(1,14) = 145.583, p < .0001$], SN mothers performed more passive nursing [$F(1,14) = 7.034, p = .0189$], and no difference between the two groups was found for arched-back nursing.

One of the most critical maternal behaviors for pup brain and behavior development is licking (Liu et al 1997; Meaney 2001). Our data show that CN pups were provided with higher levels of licking than SN pups [$F(1,14) = 78.277, p < .0001$]. Also stepping behavior was at higher levels in CN compared with SN condition [$F(1,14) = 189.084, p < .0001$], but no difference between the two groups was found for retrieving behavior.

Nesting Condition Shapes Adult Social/Aggressive Behavior

To investigate whether nesting condition exerts long-term effects on social/agonistic performance, we carried out a standard social interaction test in which two adult male mice interact socially in daily 20-min sessions for 5 consecutive days (Alleva 1993; Pietropaolo et al 2004). Nesting condition affected social investigation behaviors that are considered affiliative behaviors (Alleva 1993) (Table 1), with CN mice displaying higher frequency and longer duration of allogrooming [$F(1,42) = 17.937, p = .0001$; and $F(1,42) = 17.001, p = .0002$, respectively] and nose sniffing [$F(1,42) = 9.039, p = .0044$; and $F(1,42) = 9.391, p = .0038$, respectively], and longer duration of anogenital sniffing [$F(1,42) = 3.744, p = .0597$]. Also, social status affected social investigation behaviors (data not shown), with dominants displaying higher frequency and longer duration of allogrooming [$F(1,42) = 7.803, p = .0078$; and $F(1,42) = 6.729, p = .0130$, respectively] and lower frequency of nose sniffing [$F(1,42) = 6.418, p = .0151$]. These results suggest that CN mice showed a higher propensity to interact socially on all 5 testing days.

During the social interaction test, the two mice establish a hierarchy: the one becoming dominant, displaying a consistent amount of attacks without being attacked; and the other becoming subordinate, continuously attacked and defeated (Alleva 1993). The present results show that in SN mice, in line with previous studies (Alleva 1993; Pietropaolo et al 2004), dominants and subordinates behaved similarly on the first session; but, starting from the third session, differences began to appear. Conversely, CN mice displayed clear differences in their behavior on the first session (Figure 2). In particular, when comparing the two nesting conditions, a main effect of number of attacks was found [$F(1,42) = 10.332, p = .0025$], CN mice displaying a higher frequency of attacks than SN mice. This difference in attack behavior seems not to be due to an increased motivation to attack, as suggested by the lack of difference in the latency to the first attack (data not shown), but to the

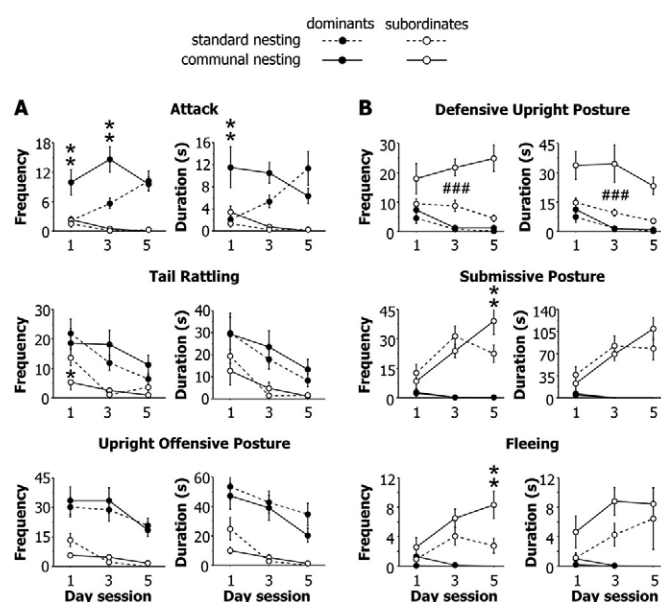


Figure 2. Social/aggressive behavior in adult male mice reared in either standard nesting (SN) or communal nesting (CN) condition. **(A)** Attack/offensive behavior. The CN dominants ($n = 10$) show a number and duration of attacks typical of dominants since the first session, whereas SN dominants ($n = 12$) need 5 days to fully behave as dominants. During the first session, SN subordinates ($n = 12$) show more tail rattling and upright offensive postures than CN subordinates ($n = 12$), suggesting that, in the first session, SN mice do not display a well defined social status. **(B)** Defensive/submissive behavior. A significant nesting condition $\times$ social status interaction was found for both frequency and duration of defensive upright postures [$F(1,42) = 15.351, p = .0003$; and $F(1,42) = 20.321, p < .0001$, respectively, both indicated by ###], with CN subordinates displaying more and longer defensive upright postures compared with SN subordinates; $*p < .05$ and $**p < .01$, versus the relative SN group (i.e., either dominants or subordinates). Data are means $\pm$ SEM.

higher number of attacks displayed by CN dominants since the first day of social interaction (Figure 1). Indeed, we found a significant nesting condition $\times$ social status $\times$ daily session interaction on both frequency and duration of attack [$F(2,84) = 64.481, p = .0110$; and $F(2,84) = 114.862, p = .0082$, respectively], CN dominants performing more attacks during the first and third sessions and longer attacks during the first session compared with SN dominants ($p < .01$). The less marked difference in attack/offensive behavior in SN compared with CN group during the first session of social interaction test concerns not only dominants but also subordinates: SN subordinates displayed tail rattling behavior, a threatening behavior usually displayed only by dominants, at levels more similar to those of dominants than CN subordinates, as shown by the higher levels of tail rattling displayed by SN subordinates compared with CN subordinates ($p < .05$; Figure 2).

To confirm whether nesting condition affects not just aggression levels but mouse ability to play social roles, we examined defensive/submissive behaviors. In this case also, mice of the CN group displayed more marked differences in their behavior according to the social role than SN mice. Nesting condition affected the frequency and the duration of defensive upright postures [$F(1,42) = 23.172$ and $F(1,42) = 26.558$, respectively, $p < .0001$], CN mice displaying more and longer defensive upright postures than SN mice at all sessions (Figure 2). Furthermore, CN subordinates displayed a more pronounced submissive behavior than SN mice, as revealed by CN subordinates displaying more submissive upright

Table 1. Propensity to Social Interaction Changes According to Early Social Experiences

	SN	CN	<i>p</i>
Allogrooming			
frequency	.70 $\pm$.40	7.65 $\pm$ 1.65	.0001
duration (sec)	2.00 $\pm$ 1.00	33.5 $\pm$ 7.4	.0002
Nose Sniffing			
frequency	23.4 $\pm$ 3.00	31.65 $\pm$ 4.50	.0044
duration (sec)	50.05 $\pm$ 8.20	87.85 $\pm$ 15.00	.0038
Anogenital Sniffing			
frequency	53.4 $\pm$ 5.65	87.1 $\pm$ 6.9	ns
duration (sec)	111.10 $\pm$ 16.75	200.00 $\pm$ 19.15	.0597

Communal nesting (CN) mice showed higher levels of social investigation behaviors than standard nesting (SN) mice. Data are mean $\pm$ SEM.

posture and fleeing, (i.e., two submissive behavioral responses) during the fifth session ($p < .01$).

No difference was found in the relative percentage of dominants versus subordinates in the CN compared with the SN mice. Standard opponents (see Methods and Materials) of CN dominants showed more crouching compared to standard opponents of SN dominants [duration (mean $\pm$ SEM): SN and CN, respectively, day 1: 1.3 ± 0.6 and 0.3 ± 0.1 ; day 3: 0.5 ± 0.2 and 20.6 ± 8.4 ; day 5: 0.8 ± 0.3 and 9.7 ± 2.6 "].

Overall, the results of the social interaction test suggest that CN, although not affecting the ability of achieving dominance, improves social competencies leading to play a social role more promptly, either dominant or subordinate.

Increased Neurotrophin Levels in CN Mice

Adult brain neurotrophin levels in rodents are decreased by adverse early experiences such as maternal separation (Cirulli et al 2003; Roceri et al 2004). We examined whether CN, a highly stimulating social environment, affected the levels of NGF and BDNF proteins at adulthood. We found that CN, opposite to adverse experiences, increased neurotrophin levels at long term (Figure 3): compared with SN mice, CN mice had higher NGF levels in the hippocampus and hypothalamus [$F(1,8) = 98.589$, $p < .0001$; and $F(1,8) = 11.977$, $p = .0086$, respectively] and higher BDNF levels in the hippocampus, striatum, and hypothalamus [$F(1,8) = 110.447$, $F(1,8) = 137.712$, and $F(1,8) = 456.573$, respectively, $p < .0001$].

Nerve growth factor levels have been reported to be acutely modulated by social status (Alleva 1993; Fiore et al 2003; Pietropaolo et al 2004). Thus, we measured NGF levels after the social interaction test and, as expected, found that subordinates had higher NGF levels than dominants in serum (data not shown), hippocampus, striatum, and hypothalamus [$F(1,18) = 5.450$, $p = .0314$; $F(1,18) = 569.521$, $p < .0001$; $F(1,18) = 402.153$, $p < .0001$; and $F(1,18) = 29.395$, $p = .0019$, respectively] (Figure 3). Furthermore, because behavioral differences between dominants and subordinates during the social interaction test were more pronounced in the CN group than in the SN group, we expected a more marked difference in NGF levels between dominants and subordinates in the CN group. Indeed, we found a significant interaction social status $\times$ nesting condition for NGF levels in hippocampus, striatum, and hypothalamus [$F(1,18) = 387.313$, $p < .0001$; $F(1,18) = 261.430$, $p < .0001$; and $F(1,18) = 7.904$, $p = .0116$, respectively]. In particular, we found that, although a difference between dominants and subordinates in SN mice was already clear in the hippocampus ($p < .05$), striatum ($p < .05$), and hypothalamus (missed statistical significance), it was more pronounced in the CN mice ($p < .01$ for all three brain areas; Figure 3). The BDNF levels were not affected by social status (Figure 3).

Discussion

The present findings indicate that CN, which characterizes the natural ecological niche of the developing mouse and represents a form of early social enrichment, markedly shapes adult social behavior. Indeed, adult CN mice, compared with SN ones, displayed a higher propensity to interact socially and more promptly achieved a defined social role. In particular, whereas mice lacking CN experience need five social encounters to fully show a behavioral profile typical of a dominant or subordinate male, mice with early social experience display a defined social role on the first encounter. These results confirm that, during early life, social

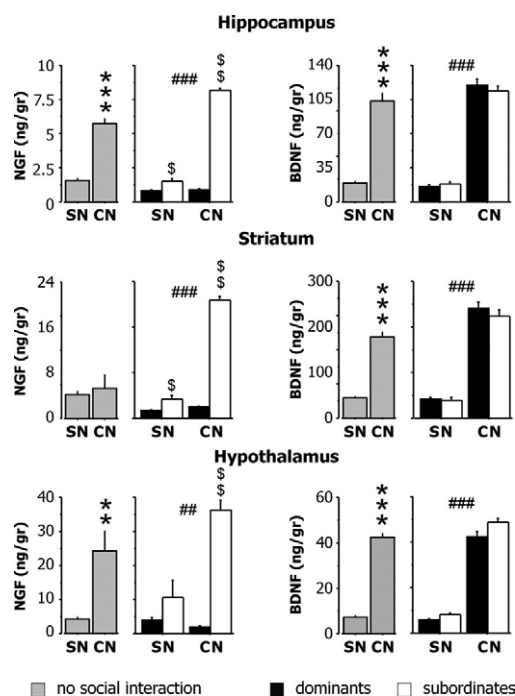


Figure 3. Brain nerve growth factor (NGF) and brain-derived neurotrophic factor (BDNF) levels are modulated according to nesting condition in mice with no adult social experience or in dominant and subordinate mice. In mice with no adult social experience (gray bars), NGF, with the exception of striatum, and BDNF levels are markedly increased in all brain areas in communal nesting (CN) compared with standard nesting (SN) group. In mice that underwent the social interaction test, subordinates (white bars) showed higher NGF levels than dominants (black bars). Furthermore, a significant interaction social status $\times$ nesting condition for NGF levels in hippocampus, striatum, and hypothalamus [$F(1,18) = 387.313$, $p < .0001$; $F(1,18) = 261.430$, $p < .0001$; and $F(1,18) = 7.904$, $p = .0116$, respectively] indicated a more pronounced difference in NGF levels in CN than SN mice. Although BDNF levels were not affected by social status, BDNF levels measured in mice that underwent the agonistic encounter further corroborated differences due to nesting condition: $**p < .01$ and $***p < .001$, respectively, versus SN mice. $^{\$}p < .05$ and $^{ss}p < .01$, respectively, versus same nesting condition dominant mice. $^{##}p < .01$ and $^{###}p < .001$, respectively, CN versus SN mice. $n = 5$ for each group. Data are means $\pm$ SEM.

behavior becomes tuned through social experience (Laviola and Terranova 1998; Mendl and Paul 1990a, 1990b). Furthermore, in association with changes in social behavior, CN mice show higher NGF levels, which are further modulated according to social/aggressive behavior, and higher BDNF levels.

In rodents, the mother is the major source of early experiences for the developing pup that depends on her for most of its needs (Hofer 1984). We show that in the CN condition, compared with SN, augmented levels of maternal behavior, particularly nursing and licking, are displayed by the mothers. We hypothesize that these higher levels of maternal behavior account, at least in part, for the behavioral differences between mice exposed to the two rearing conditions. Indeed, it has been widely shown that variations in maternal behavior produce long-term modifications in offspring behavior (Cirulli et al 2003; Cushing and Kramer 2005; Meaney 2001), such as changes in anxiety-like behavior (Caldji et al 1998), spatial learning, and memory performance (Liu et al 2000). Furthermore, in line with our results, decreased maternal behavior has been associated with reduced adult social behavior (Macri and Laviola 2004; Mintz et al 2005), suggesting a direct relationship

between levels of maternal behavior and propensity to interact socially at adulthood.

Growing up with a high number of littermates, as in the CN condition, might represent an additional factor affecting mouse behavioral development. Indeed, the relevance of early peer-to-peer interactions in the emergence of differences in adult social behavior has been demonstrated in rodents (Bautista et al, *in press*; Hudson 2005; Laviola and Alleva 1995; Terranova and Laviola 1995). Also, studies on rhesus monkeys corroborate this point (Suomi 2005). Peer-reared infant rhesus monkeys tend to engage in play with age-mates more frequently than those reared in mother-infant dyads, because having been reared solely by their mothers, who are far less likely than infants to engage in playful behavior, might have deprived them of appropriate social experience (Suomi 1979, 2005). In humans, the relevance of early social environment in shaping adult behavior has been widely reported (Canetti et al 1997; Parker 1981; Russek and Schwartz 1997; Rutter et al 1997). In particular, those children who experience an aversive social environment are at risk for the emergence of difficulties in social/aggressive interaction later in life (Dodge et al 1995; Rogosch et al 1995) that eventually lead to further social problems due to peer rejection (Bolger and Patterson 2001).

Here we also show that the differences in social/aggressive behavior between adult CN and SN mice are associated with markedly different neurotrophin levels. Indeed, we found that NGF levels were increased approximately five-fold in the hippocampus and hypothalamus of CN compared with SN mice and were further affected by social status, with subordinates showing higher NGF levels compared with dominants. It is interesting to notice that the more marked difference in behavior between dominants and subordinates in the CN compared with SN mice is paralleled by a more marked difference in NGF levels. This result corroborates the reported association between dominant or subordinate behavioral profile and NGF levels (Avitsur et al 2001; Branchi et al 2004; Pietropaolo et al 2004), further pointing out the crucial role of social experiences in modulating neurotrophin levels. Indeed, NGF levels are reportedly affected by social factors, such as psychosocial stress, both in laboratory animals (Aloe et al 1986; Avitsur et al 2001; Branchi et al 2004; Lakshmanan 1986; Maestripieri et al 1990) as well as in humans (Hadjicostantinou et al 2001). The NGF increase found in subordinate mice might play a role in increasing brain plasticity for the response to social challenges faced by subordinate mice (Alleva and Aloe 1989; Branchi et al 2004; Sapolsky 2005). Brain-derived neurotrophic factor levels are increased approximately eight-fold in CN compared with SN mice in all the adult brain areas analyzed, indicating that the levels of this neurotrophin are highly modulated by early social experiences. Our results are in line with previous results showing that adverse early experiences (i.e., maternal separation) lead to a reduction of BDNF levels at adulthood (Roceri et al 2002, 2004). Other studies, although confirming a direct relationship between maternal behavior and BDNF levels, report only short-term effects of variation of maternal care on neurotrophin levels (Liu et al 2000); however, differences in the early social stimulation provided to pups might account for this apparent discrepancy.

We hypothesize that the increase in brain NGF and BDNF levels as a consequence of early social enrichment might be important in shaping brain plasticity to face a highly demanding social life at adulthood. Indeed, social interactions have been reported to affect neurotrophin levels (Branchi et al 2004). The mouse lives in a variety of ecological conditions, each one with varied environmental characteristics, such as different population densities. Furthermore, because it has a short life span, the mouse usually spends its

adult life in an environment similar to the one in which it was born. Thus, shaping brain function in early life might be of adaptive value to the adult (Hofer 1983). In this case, high neurotrophin levels and thus a highly plastic brain might be of adaptive value to the adult that has to face many social interactions. A similar epigenetic mechanism has already been suggested for the development of other neurophysiological responses, such as the activity of the hypothalamic–pituitary–adrenal axis in response to stress (Liu et al 1997; Meaney 2001). It is also interesting to notice that the increase in BDNF levels found in CN compared with SN mice is independent of social status. Therefore, because stressful events reportedly lead to a reduction of BDNF levels (Smith et al 1995; Ueyama et al 1997), it seems that CN dominants and subordinates have a similar stress response to social challenge, in agreement with behavioral results showing that both groups display improved competencies in facing social interactions. This is in line with the fact that the levels of stress associated with being dominant or subordinate change according to social context and the individual's ability to face social challenges (Goymann and Wingfield 2004; Sapolsky 2005). Further studies are warranted to assess physiological differences in coping response to stress in SN and CN mice.

In summary, CN enhances propensity to interact socially, shapes social competencies, and increases neurotrophin levels at adulthood. These findings are consistent with the hypothesis that alterations in the early social environment, which involve changes in parental care, affect brain and behavior development (Bowlby 1968). Furthermore, because impairments in social behavior and altered neurotrophin levels, particularly decreased BDNF levels, have been associated with several psychiatric diseases, including depression and schizophrenia (Castren 2005; First 1996; Nawa et al 2000; Schumacher et al 2005), our results suggest that that early social environment might affect the likelihood in susceptible individuals to develop selected symptoms of psychopathology. Finally, the present findings indicate CN as an alternative experimental strategy for the study of long-term effects of early experiences on brain and behavior that, in contrast to the adverse experience of maternal deprivation, provides a highly stimulating social environment to the developing rodent.

This study was supported by project NIH-ISS Rif. OF14 to EA and project ISS Stem cells N. 23 to LA. We warmly thank Francesca Cirulli and Gianni Laviola for critical reading and Jantine Sietzema, Nadia Francia, and Giovanni Dominici for technical support.

- Alleva E (1993): Assessment of aggressive behaviour in rodents. In: Conn MP, editor. *Methods in Neuroscience. Paradigms for the Study of Behavior*, Vol 14. New York: Academic Press, 111–137.
- Alleva E, Aloe L (1989): Physiological roles of nerve growth factor in adult rodents: A biobehavioral perspective. *Int J Comp Psychol* 2:147–163.
- Aloe L, Alleva E, Bohm A, Levi-Montalcini R (1986): Aggressive behavior induces release of nerve growth factor from mouse salivary gland into the bloodstream. *Proc Natl Acad Sci U S A* 83:6184–6187.
- Avitsur R, Stark JL, Sheridan JF (2001): Social stress induces glucocorticoid resistance in subordinate animals. *Horm Behav* 39:247–257.
- Bautista A, Martínez-Gómez M, Hudson R (in press): Mother-young and within-litter relations in the European rabbit *Oryctolagus cuniculus*. In: Célio Alves P, Hackländer K, editors. *Lagomorph Biology: Evolution, Ecology and Conservation*.
- Bolger KE, Patterson CJ (2001): Developmental pathways from child maltreatment to peer rejection. *Child Dev* 72:549–568.
- Bowlby J (1968): Effects on behaviour of disruption of an affectional bond. *Eugen Soc Symp* 4:94–108.
- Branchi I, Francia N, Alleva E (2004): Epigenetic control of neurobehavioural plasticity: The role of neurotrophins. *Behav Pharmacol* 15:353–362.

- Caldji C, Tannenbaum B, Sharma S, Francis D, Plotsky PM, Meaney MJ (1998): Maternal care during infancy regulates the development of neural systems mediating the expression of fearfulness in the rat. *Proc Natl Acad Sci U S A* 95:5335–5340.
- Canetti L, Bachar E, Galili-Weisstub E, De-Nour AK, Shalev AY (1997): Parental bonding and mental health in adolescence. *Adolescence* 32:381–394.
- Castren E (2005): Is mood chemistry? *Nat Rev Neurosci* 6:241–246.
- Cirulli F, Berry A, Alleva E (2003): Early disruption of the mother-infant relationship: Effects on brain plasticity and implications for psychopathology. *Neurosci Biobehav Rev* 27:73–82.
- Cushing BS, Kramer KM (2005): Mechanisms underlying epigenetic effects of early social experience: The role of neuropeptides and steroids. *Neurosci Biobehav Rev* 29:1089–1105.
- Dodge KA, Pettit GS, Bates JE, Valente E (1995): Social information-processing patterns partially mediate the effect of early physical abuse on later conduct problems. *J Abnorm Psychol* 104:632–643.
- Dube SR, Anda RF, Felitti VJ, Chapman DP, Williamson DF, Giles WH (2001): Childhood abuse, household dysfunction, and the risk of attempted suicide throughout the life span: Findings from the Adverse Childhood Experiences Study. *JAMA* 286:3089–3096.
- Fiore M, Amendola T, Triaca V, Tirassa P, Alleva E, Aloe L (2003): Agonistic encounters in aged male mouse potentiate the expression of endogenous brain NGF and BDNF: Possible implication for brain progenitor cells' activation. *Eur J Neurosci* 17:1455–1464.
- First MB (1996): *DSM-IV. Diagnostic and statistical manual of mental disorders*, 4th ed. Amsterdam: American Psychiatric Association.
- Goymann W, Wingfield JC (2004): Allostatic load, social status and stress hormones: The costs of social status matter. *Anim Behav* 67:591–602.
- Hadjiconstantinou M, McGuire L, Duchemin AM, Laskowski B, Kiecolt-Glaser J, Glaser R (2001): Changes in plasma nerve growth factor levels in older adults associated with chronic stress. *J Neuroimmunol* 116:102–106.
- Harlow HF (1958): The nature of love. *Am Psychol* 13:537–685.
- Hayes LD (2000): To nest communally or not to nest communally: A review of rodent communal nesting and nursing. *Anim Behav* 59:677–688.
- Hofer MA (1983): The mother-infant interaction as a regulator of infant physiology and behavior. In: Rosenblum LA, Moltz H, editors. *Symbiosis in Parent-Offspring Interactions*, Vol 104. New York: Plenum Press, 61–75.
- Hofer MA (1984): Relationships as regulators: A psychobiologic perspective on bereavement. *Psychosom Med* 46:183–197.
- Hudson R (2005): The contribution of sibling relations to the emergence of individual behavioural phenotypes. International Conference of Ethology Conference Proceedings, August 20–27, 2005. Budapest, Hungary: Keszult a Codex Print Nyomdaban, pp. 6.
- Lakshmanan J (1986): Aggressive behavior in adult male mice elevates serum nerve growth factor levels. *Am J Physiol* 250:E386–E392.
- Laviola G, Alleva E (1995): Sibling effects on the behavior of infant mouse litters (*Mus domesticus*). *J Comp Psychol* 109:68–75.
- Laviola G, Terranova ML (1998): The developmental psychobiology of behavioural plasticity in mice: The role of social experiences in the family unit. *Neurosci Biobehav Rev* 23:197–213.
- Levine S (1957): Infantile experience and resistance to physiological stress. *Science* 126:405.
- Liu D, Diorio J, Day JC, Francis DD, Meaney MJ (2000): Maternal care, hippocampal synaptogenesis and cognitive development in rats. *Nat Neurosci* 3:799–806.
- Liu D, Diorio J, Tannenbaum B, Caldji C, Francis D, Freedman A, et al (1997): Maternal care, hippocampal glucocorticoid receptors, and hypothalamic-pituitary-adrenal responses to stress. *Science* 277:1659–1662.
- Macri S, Laviola G (2004): Single episode of maternal deprivation and adult depressive profile in mice: Interaction with cannabinoid exposure during adolescence. *Behav Brain Res* 154:231–238.
- Maestripietri D, De Simone R, Aloe L, Alleva E (1990): Social status and nerve growth factor serum levels after agonistic encounters in mice. *Physiol Behav* 47:161–164.
- McAllister AK, Katz LC, Lo DC (1999): Neurotrophins and synaptic plasticity. *Annu Rev Neurosci* 22:295–318.
- McEwen BS (1998): Protective and damaging effects of stress mediators. *N Engl J Med* 338:171–179.
- Meaney MJ (2001): Maternal care, gene expression, and the transmission of individual differences in stress reactivity across generations. *Annu Rev Neurosci* 24:1161–1192.
- Mendl M, Paul ES (1990a): Parental care, sibling relationships and the development of aggressive behaviour in two lines of wild house mice. *Behavior* 116:90–108.
- Mendl M, Paul ES (1990b): Litter composition affects parental care, offspring growth, and the development of aggressive behaviour in wild house mice. *Behavior* 116:11–41.
- Mintz M, Ruedi-Bettschen D, Feldon J, Pryce CR (2005): Early social and physical deprivation leads to reduced social motivation in adulthood in Wistar rats. *Behav Brain Res* 156:311–320.
- Nawa H, Takahashi M, Patterson PH (2000): Cytokine and growth factor involvement in schizophrenia—support for the developmental model. *Mol Psychiatry* 5:594–603.
- Parker G (1981): Parental representations of patients with anxiety. *Acta Psychiatr Scand* 63:33–36.
- Pietropaolo S, Branchi I, Cirulli F, Chiarotti F, Aloe L, Alleva E (2004): Long-term effects of the periadolescent environment on exploratory activity and aggressive behaviour in mice: Social versus physical enrichment. *Physiol Behav* 81:443–453.
- Pollak SD (2003): Experience-dependent affective learning and risk for psychopathology in children. *Ann N Y Acad Sci* 1008:102–111.
- Roceri M, Cirulli F, Pessina C, Peretto P, Racagni G, Riva MA (2004): Post-natal repeated maternal deprivation produces age-dependent changes of brain-derived neurotrophic factor expression in selected rat brain regions. *Biol Psychiatry* 55:708–714.
- Roceri M, Hendriks W, Racagni G, Ellenbroek BA, Riva MA (2002): Early maternal deprivation reduces the expression of BDNF and NMDA receptor subunits in rat hippocampus. *Mol Psychiatry* 7:609–616.
- Rogosch FA, Cicchetti D, Laber J (1995): The role of child maltreatment in early deviations in cognitive and affective processing abilities and later peer relationship problems. *Dev Psychopathol* 7:591–609.
- Rosenzweig MR (2003): Effects of differential experience on the brain and behavior. *Dev Neuropsychol* 24:523–540.
- Russek LG, Schwartz GE (1997): Feelings of parental care predict health status in midlife: A 35 year follow-up of the Harvard Mastery of Stress Study. *J Behav Med* 20:1–13.
- Rutter M, Dunn J, Plomin R, Simonoff E, Pickles A, Maughan B, et al (1997): Integrating nature and nurture: Implications of person-environment correlations and interactions for developmental psychopathology. *Dev Psychopathol* 9:335–364.
- Sale A, Putignano E, Cancedda L, et al (2004): Enriched environment and acceleration of visual system development. *Neuropharmacology* 47:649–660.
- Sapolsky RM (2005): The influence of social hierarchy on primate health. *Science* 308:648–652.
- Sayler A, Salmon M (1971): An ethological analysis of communal nursing by the house mouse (*Mus musculus*). *Behavior* 40:60–85.
- Schumacher J, Jamra RA, Becker T, Ohlraun S, Klopp N, Binder EB, et al (2005): Evidence for a relationship between genetic variants at the brain-derived neurotrophic factor (BDNF) locus and major depression. *Biol Psychiatry* 2005 58:307–314.
- Smith MA, Makino S, Kvetnansky R, Post RM (1995): Stress and glucocorticoids affect the expression of brain-derived neurotrophic factor and neurotrophin-3 mRNAs in the hippocampus. *J Neurosci* 15:1768–1777.
- Suomi SJ (1979): Peers, play and primary prevention in primates. In: Kent M, Rolf J, editors. *Primary Prevention of Psychopathology: Social Competence in Children*. Hanover, New Hampshire: Press of New England, 127–149.
- Suomi SJ (2005): Mother-infant attachment, peer relationships, and the development of social networks in rhesus monkeys. *Hum Dev* 48:67–79.
- Terranova ML, Laviola G (1995): Individual differences in mouse behavioural development: Effects of precocious weaning and ongoing sexual segregation. *Anim Behav* 50:1261–1271.
- Trickett PK, McBride-Chang C (1995): The developmental impact of different forms of child abuse and neglect. *Dev Rev* 15:311–337.
- Ueyama T, Kawai Y, Nemoto K, Sekimoto M, Tone S, Senba E (1997): Immobilization stress reduced the expression of neurotrophins and their receptors in the rat brain. *Neurosci Res* 28:103–110.